

Unit 5, Lesson 3: Solving Inequalities – Part 2

Benchmark

MA.B.AR.2.2 Solve two-step linear inequalities in one variable and represent solutions algebraically and graphically.

Learning Target

- ☐ I can solve inequalities and represent the solutions on a number line.

5.3.1 Warm-Up: Creating Inequalities

1. Using the integers -8 to 8 at most one time each, create an inequality with solutions of $x > \frac{2}{3}$. There is more than one solution.

$x <$

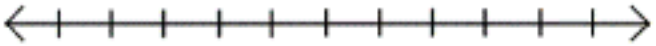
5.3.2 Try It: Card Match

1. Work with your partner to match the provided inequality and solution cards to each graph in the table.

Graph	Inequality	Solution

5.3.3 Your Turn!

1. Solve each inequality and represent the solutions on the number line.

a. $4x - 5 \geq -25$ 	b. $-1.2 < \frac{x-4}{5}$ 
c. $0.2n + 1 \geq -1\frac{3}{5}$ 	d. $-\frac{1}{2}(-3a + 8) \leq -15$ 
e. $0.4(6 - 5.2x) > -8$ 	f. $\frac{2}{3}a - 5 > 1$ 
g. $-3 \leq 4.5 - b$ 	h. $-11 - \frac{y}{4} < -9$ 

5.3.4 Collaboration: Solving Real-World Inequalities

Work with your partner to determine the meaning of each inequality in terms of the given situation.

1. Sasha wants to save at least \$300 to purchase a new gaming system. She asked Victor to help her figure out how much she would need to save each month if she has already saved \$50 and would like to buy a system in 6 months. Victor wrote the inequality $6m + 50 \geq 300$, where m represents the dollar amount Sasha must save each month.
- a. Discuss what the values 6 and 50 represent in the inequality.
- b. In terms of the situation, what are the limitations on the possible values of m ?
- c. Solve the inequality.

- d. Explain what the solution means in terms of the situation.
2. Sasha received \$100 for her birthday and hopes to buy 4 video games for her gaming system. Each game usually costs \$40, but she’s hoping she can find the games on sale this week. She writes the inequality $4(40 - d) \leq 100$, where d represents the dollar amount of the discount on the video games.

a. What does the expression $40 - d$ represent?

b. Solve the inequality.

c. Explain what the solution means in terms of the situation.

d. Do you think it is likely that Sasha will be able to purchase the 4 video games she wants?

5.3.5 Wrap-Up: Growth Mindset

1. Reflect on your work today and describe your favorite mistake that allowed you to acquire new knowledge, experience, or skills.

Means
I
Start
To
Acquire
Knowledge
Experience
Skills

